

Curriculum Vitae - Prateek Dwivedi

Education

Indian Institute of Science Education and Research (IISER), Pune <i>BS-MS Dual Degree: Mathematics</i>	2020 – 2025
Technische Universität (TU) Ilmenau, Germany <i>Visiting Research: Mathematics (Coding Theory)</i>	2024 – 2025
IIT Delhi <i>PhD: Mathematics (Combinatorics); discontinued for health reasons</i>	2025
IIT Madras <i>PhD (SIDiS) at Inteface Lab</i>	2026 – Present

Research Experience

Statistical Modeling of Gene Expression <i>IISER Pune (Prof. Leelavati Narlikar)</i>	July 2022 – Dec 2022
Built statistical models to predict gene expression from promoter sequence data	
Stochastic Processes: Matingale Reading Project <i>IISER Pune (Prof. Anup Biswas)</i>	January 2023 – April 2023
Read an introduction to martingale theory from the book “Stochastic Calculus and Applications” by Cohen and Elliot, and an alternative interpretation of Doob’s Martingale Inequality from “A Trajectorial Interpretation of Doob’s Martingale Inequalities” by Acciaio, Beiglböck, Penkner, Schachermayer, and Temme.	
Coding Theory Project: Bounds on Asymptotic Rate Function <i>IISER Pune (Prof. Krishna Kaipa)</i>	May 2023 – Dec 2023
Studied bounds on asymptotic rate function and produced alternative proofs of the MacWilliams identity for block codes	
Finite Geometry: Classification of Conic Pencils <i>IISER Pune (Prof. Krishna Kaipa)</i>	Dec 2023 – Mar 2024
Classified pencils of conics in $\text{PG}(2, q)$ under the action of $\text{PGL}_3(q)$	
Convolutional Coding Theory: Thesis Research <i>TU Ilmenau (Prof. Julia Lieb) & IISER Pune (Prof. Krishna Kaipa)</i>	Apr 2024 – Mar 2025
Title: “Convolutional Codes: MacWilliams Identity and Hulls.” - Read papers on the MacWilliams identity for convolutional codes by Prof. Heide Gluesing-Luerssen and Gert Schneider, and did a literature review of the same. Original contribution: extended LCD codes and code hulls to the convolutional setting and proved a hull-dimension formula using the rank of the generator matrix, and gave an explicit construction for convolutional LCD codes using Lagrange’s four-square theorem, as an extension to a similar construction for block LCD codes.	
Topological Combinatorics: Arrangements of Curves <i>IIT Delhi (Prof. Rajiv Raman)</i>	
Investigating problems in topological combinatorics related to arrangements of curves and Levi’s Extension Theorem for non-piercing curves.	
Interpretable Attention Models for Dynamics <i>IIT Madras (Interface Lab - Prof. Ganga S Pratsath)</i>	

Additional Information

- Speaker, Coding Theory: SURC 2023 (Summer Undergraduate Research Conference): Azim Premji University
- Attendee, Indo-European Conference on Mathematics (IECM) January 2026, IISER Pune and SPPU